Haosen Cao

CURRICULUM VITAE *Last updated: September 2026*

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EDUCATION

University of Southern California

Master of Science, Computer Science | **GPA: 3.73 / 4.0** *Jan 2025 – Dec 2026 (expected) · Los Angeles, CA*

Relevant coursework: Information Retrieval and Web Search Engines (CSCI 572), Database Systems (CSCI 585), Foundations of Modern Data Management and Processing (CSCI 543), Analysis of Algorithms (CSCI 570), Web Technologies (CSCI 571), Software Engineering (CSCI 577a).

University of California, Davis

Bachelor of Science, Computer Science | Minor in Japanese | **GPA: 3.54 / 4.0** *Sep 2020 – Jun 2024 · Davis, CA*

Relevant coursework: Machine Learning (ECS 171), Algorithm Design & Analysis (ECS 122A), Operating Systems (ECS 150), Computer Architecture (ECS 154A), Programming Languages (ECS 140A), Theory of Computation (ECS 120), Information Interfaces (ECS 163).

RESEARCH INTERESTS

Information Retrieval & Ranking Systems · Evaluation, Reliability & Attribution of LLM-Based Systems · Database Systems & Query Processing

PUBLICATIONS & PREPRINTS

Yi Nian, **Haosen Cao**, Shenzhe Zhu, Henry Peng Zou, Qingqing Luan, Yudi Zhang, Yue Zhao. "When Only the Final Text Survives: Implicit Execution Tracing for Multi-Agent Auditing." *arXiv:2603.17445 [cs.AI, cs.CL]*, 2026. **Under review.**
arxiv.org/abs/2603.17445

- Proposed Implicit Execution Tracing (IET), a provenance-by-design framework embedding agent-specific, key-conditioned statistical signals into token generation, so that the final text becomes a self-verifying provenance record: an offline auditor holding a registry of per-agent keys recovers segment boundaries and per-segment attribution from that text alone, without execution logs or private traces.
- *Contributions:* the multi-agent attribution experiments and benchmarking against watermarking baselines; robustness evaluation under identity removal, boundary corruption, and privacy-preserving redaction; ablation studies on attribution accuracy, segment overlap, and transition-structure recovery.

RESEARCH EXPERIENCE

University of Southern California, Viterbi School of Engineering

Research Assistant · Advised by Prof. Yue Zhao and Prof. Mengyuan Li *Jul 2026 – Present · Los Angeles, CA*

Retrieval and ranking for two-sided matching, on the same production system as the industry work below; the entries here concern architecture, methodology, and controlled study.

EMPLOYER-SIDE MATCHING ARCHITECTURE

- Designed a three-layer architecture for job-to-candidate matching — hard filtering, coarse retrieval over a whole-profile embedding, then per-field cosine scoring with weighted summation — in which both sides decompose into field blocks under a shared template and the ranking layer emits a per-field score profile rather than a scalar, rendering every ranking decision attributable to the passage that produced it.
- Established per-field standardization as a precondition for interpretable weighting, since a field's influence on the ordering scales with the product of its weight and its own dispersion; statistics are estimated offline over the full library, preserving comparability across queries and admitting incremental extension of a result set.
- Rejected rank-reciprocal fusion within the ranking layer on the grounds that its premise — combining scorers without a common scale — does not obtain there, and that rank substitution discards effect magnitude.

- Formalized missing-value semantics as renormalization over observed fields rather than zero-imputation, distinguishing absence of requirement on the posting side from absence of evidence on the candidate side.
- Constrained the filtering stage to predicates expressible within the retrieval query, separating non-negotiable qualifications from relaxable ranges, and excluded health and physical-condition attributes from the schema as a compliance boundary.
- Specified the resulting score matrix as the interface for two downstream uses: attribution of employer preference into weight adjustments, and per-candidate gap identification for targeted follow-up.

MULTIMODAL ASSESSMENT OF ASYNCHRONOUS INTERVIEWS

- Designed and implemented a staged pipeline reducing a single recording — camera and screen capture, without operating-system instrumentation — to timestamped structured records in which every assertion carries a verifiable source in the transcript or an identified frame.
- Adopted dual-granularity segmentation on the observation that the resolution at which a citation is verifiable differs from that at which a description is coherent, and specified requirement coverage, citation verifiability, and event-to-frame traceability as the evaluation criteria the pipeline would have to meet.
- Bounded the system's claim to verifying consistency between assertion and demonstrated behavior rather than grading performance, treating model judgments of fine-grained quality as unvalidated against human annotation and therefore outside what the system asserts.
- Surveyed the legal and adversarial boundary of the approach: provider constraints on workplace affect inference, biometric-privacy exposure, and prompt injection delivered through shared screen content.

CONTROLLED STUDY OF THE RETRIEVAL EMBEDDING LAYER

- Conducted controlled ablations, each with a length- and format-matched control arm, to determine which structured fields carry signal in the indexed representation; a rule-based extractor recovered most of the benefit of an LLM-based one without additional latency.
- Demonstrated that query and document embeddings drawn from different models cannot be mixed — retrieval degrades to chance while surface indicators remain nominal — establishing atomicity as a requirement for model migration.
- Fine-tuning the retrieval embedding model on in-domain data.

Figwork

Research Intern · Retrieval & Ranking *Jun 2026 — Present · Remote*

Retrieval and ranking layer of a deployed LLM job-matching system serving a browser-extension product: Python with FastAPI and SQLAlchemy over PostgreSQL with pgvector HNSW indexes, Vertex AI embedding and generation models, on Compute Engine, Cloud Run, and Cloud SQL.

- Reconstructed the system's architecture from source, live services, and database state following the departure of its author, correcting inverted environment records in the inherited documentation.
- Characterized and repaired a class of silent ranking failures — insufficient approximate-nearest-neighbor search width, rank fusion counting configured rather than productive retrieval routes, score filters that never reorder, token-boundary truncation, and a configuration path admitting query and document embeddings from disjoint spaces — each degrading result quality while surface metrics and confidence scores remained nominal.
- Designed and deployed a multi-field candidate filtering layer separating predicates admissible within the retrieval query from those requiring soft weighting, with fail-open semantics under missing source data.
- Established the team's evaluation infrastructure: a corpus of authentic resumes, a cross-vendor judge ensemble chosen against shared model bias, reproducible relevance scorecards, and a full-route post-deployment suite adopted as the release gate.
- Instituted a same-session A/B protocol after measuring cross-day judge drift exceeding the effects under test, and calibrated the ensemble's reliability to bound the granularity of admissible conclusions.
- Reversed a standing assumption concerning the lexical retrieval path through controlled live experiments, attributing its negative contribution to the fragmentation of a controlled vocabulary under full-text indexing.

Huazhong University of Science and Technology

Undergraduate Research Assistant · School of Computer Science *Jul 2023 — Aug 2023 · Wuhan, China*

- Investigated the NP-hard k-coloring problem in C, comparing local search, tabu search, and hybrid evolutionary metaheuristics on large sparse graphs.

- Obtained 500-color solutions for graphs of 10,000 edges within 60 seconds on average — a threefold improvement over the baseline — through tuning of neighborhood structure and candidate-list strategy.

Mobvoi (出门问问)

Algorithm Engineer · Text-to-Image Generation Group *Aug 2024 – Dec 2024 · Suzhou, China*

- Trained and fine-tuned a family of style-specific text-to-image diffusion models, together with the scoring models used to select among their checkpoints.
- Benchmarked three text-to-image diffusion models across aesthetic quality, prompt fidelity, style consistency, and diversity, and formulated a composite metric integrating FID, CLIP score, aesthetic scoring, and prompt consistency for checkpoint selection.
- Built an internal annotation platform that materially increased the team's daily yield of validated samples, supporting checkpoint evaluation and style-conditioned data curation.

SELECTED PROJECTS

Adaptive Cuckoo Hash Join in DuckDB

C++ · USC CSCI 543 – Foundations of Modern Data Management and Processing *Jan 2026 – Apr 2026*

- Implemented a selectable cuckoo hash join backend within DuckDB's execution engine, extended to an adaptive hybrid with BFS relocation, block-structured buckets, stash overflow handling, and fallback to linear probing under load.
- Validated against the Join Order Benchmark and TPC-DS, characterizing the trade-off between probing cost, relocation overhead, and memory locality.

California EV Adoption Data Visualization

JavaScript, D3.js · UC Davis ECS 163 *Jan 2024 – Mar 2024*

Narrative visualization of California EV adoption from 2010 to 2023 through brushable timelines, choropleth maps, and linked scatterplots.

WS Toolbox (cardtoolbox.org)

React.js, Node.js, Express, MongoDB *Apr 2025 – Present*

Full-stack management tool for the trading card game Weiss Schwarz: a RESTful backend over 58,000+ card records in MongoDB, a responsive React.js interface, and cloud deployment with environment isolation.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, R, HTML/CSS

Systems & Databases: PostgreSQL (pgvector, HNSW), DuckDB, MongoDB, SQLite, Linux, Docker, CMake, Git

ML & Data: PyTorch, scikit-learn, NumPy, pandas, D3.js

Methods: embedding-based retrieval, rank fusion, LLM-as-judge evaluation and calibration, controlled A/B experiment design, ablation studies

Frameworks & Cloud: FastAPI, SQLAlchemy, React.js, Node.js, Express; Google Cloud Platform (Vertex AI, Compute Engine, Cloud Run, Cloud SQL)

Languages: English (Fluent), Chinese (Native), Japanese (Basic)

LEADERSHIP & SERVICE

UC Davis Picnic Day 2022

Technical Vice Director *Feb 2022 – May 2022 · Davis, CA*

Designed the official event website and coordinated 50+ audio equipment sets across five departments for on-site and presentation venues, for UC Davis's annual open-house event (~80,000 attendees).